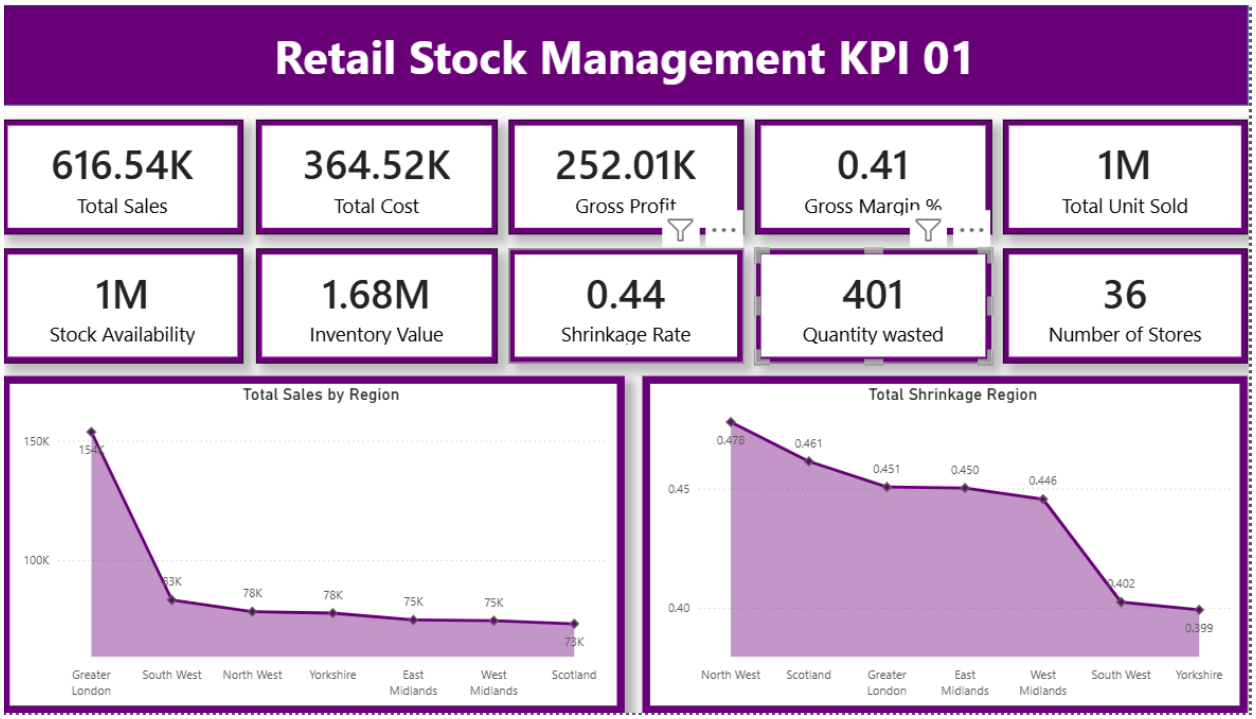
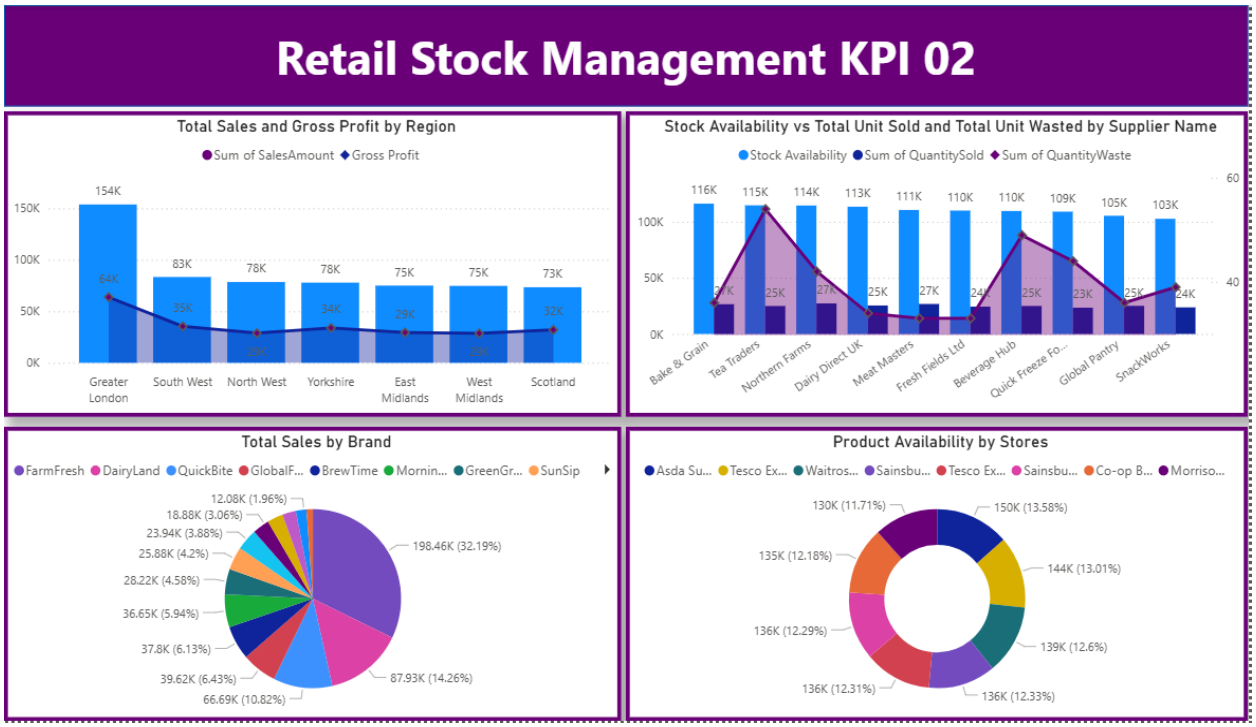
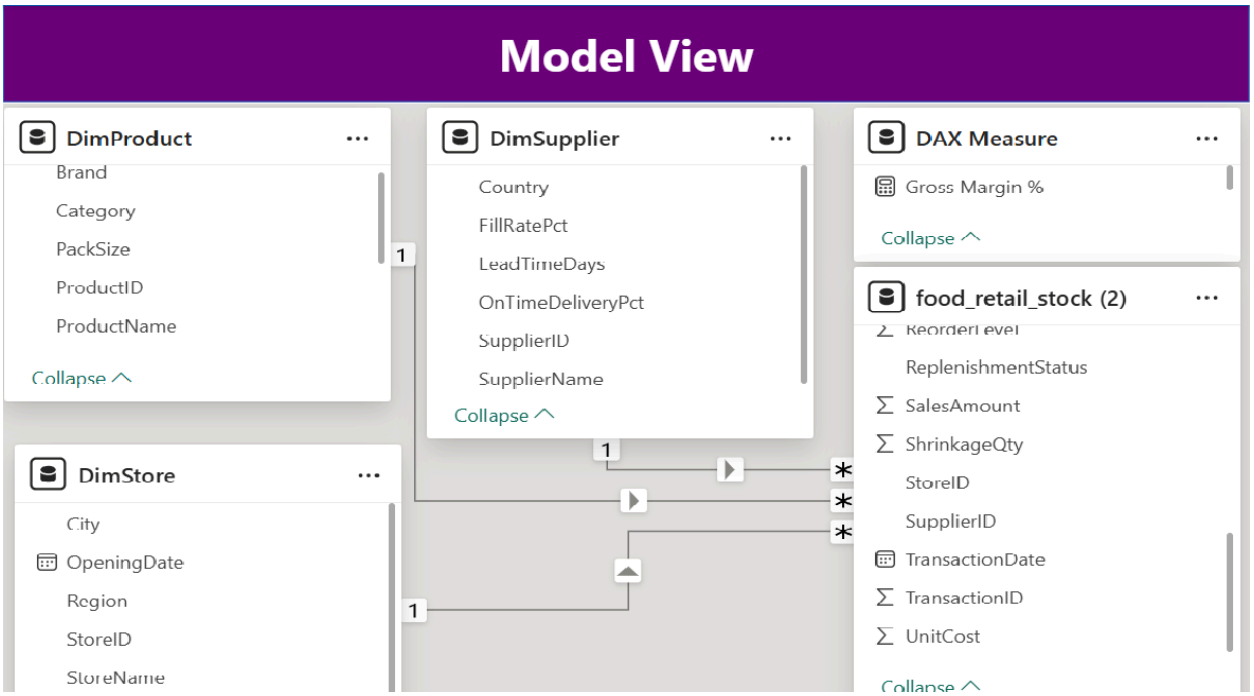
Food retail stock management dataset for SQL Server and Power BI with approximately 10,000 rows. The dataset contains data quality issues such as duplicates, missing values, inconsistent category names, invalid dates, negative quantities, and outliers. The data cleaning process was carried out using SQL Server, while the dashboard was created in Power BI using DAX measures and other useful visualisations







Folder Structure

```text
Food-Retail-Stock-Management/
│
├── data/
│   ├── raw/
│   └── cleaned/
│
├── sql/
│   ├── create_tables.sql
│   ├── cleaning_queries.sql
│   └── analysis_queries.sql
│
├── powerbi/
│   └── dashboard.pbix
│
├── images/
│   ├── dashboard_page1.png
│   └── dashboard_page2.png
│
└── README.md
```

Food Retail Stock Management Dashboard

A **Food Retail Stock Management Analytics Project** built using **SQL Server** for data preparation and cleaning, and **Power BI** for interactive dashboard reporting. The project uses a dataset of approximately **10,000 rows** designed to simulate real-world retail inventory and sales operations.

Project Overview

This project focuses on analysing food retail stock data to support inventory control, sales monitoring, and operational decision-making. The sample dataset was intentionally created with a realistic business structure and includes common data quality issues so the cleaning process can be practised and demonstrated.

The workflow for this project includes:

- Generating and structuring a food retail dataset.
- Importing and analysing the data in **SQL Server**.
- Cleaning data quality issues using SQL queries.
- Building an interactive **Power BI** dashboard.
- Creating meaningful **DAX measures** and visuals for business insights.

Project Objective

The main objective of this project is to simulate a real-world **food retail stock management system** and create an analytical dashboard that helps answer key business questions, such as:

- Which products are running low in stock?
- Which items are out of stock?
- Which categories perform best?
- What is the total inventory value?
- Which suppliers or locations need attention?
- How does stock movement affect sales and replenishment?

Tools and Technologies Used

- **SQL Server** – for data storage, transformation, and cleaning.
- **Power BI** – for dashboard creation and data visualisation.

- **DAX** – for calculated measures and business KPIs.
- **Power Query Editor** – for additional data preparation and shaping.
- **GitHub** – for project version control and documentation.

Dataset Summary

The dataset contains approximately **10,000 records** representing food retail stock and sales activity. It is designed to support stock management analysis and Power BI reporting.

Dataset Features

The dataset includes fields such as:

- Transaction ID
- Transaction Date
- Store ID
- Store Name
- Product ID
- Product Name
- Category
- Supplier ID
- Supplier Name
- Opening Stock
- Quantity Sold
- Quantity Received
- Closing Stock
- Reorder Level
- Unit Cost
- Unit Price
- Sales Amount
- Cost Amount
- Expiry Date
- Batch Number
- City
- Region
- Payment Type

Data Quality Issues Included

To make the project more realistic and useful for learning data cleaning, the dataset includes several intentional data issues:

- Duplicate records.
- Missing values in selected columns.
- Inconsistent category names.
- Invalid dates.
- Negative quantities.
- Outlier values in the price and stock columns.
- Mixed text casing.
- Extra spaces in text fields.
- Null or incomplete supplier and store values.

These issues were cleaned using **SQL Server**, making the dataset more reliable for analysis and dashboard creation.

Data Cleaning Process

The data cleaning stage was completed in SQL Server before loading the final dataset into Power BI. The following cleaning tasks were performed:

- Removed duplicate rows.
- Handled missing values.
- Standardised category and product names.
- Corrected invalid date values.
- Treated negative quantities where necessary.
- Identified and reviewed outliers.
- Trimmed extra spaces.
- Standardised text casing.
- Validated stock and sales-related fields.

This step ensured that the final dataset was accurate enough for meaningful reporting and KPI calculation.

Dashboard Overview

The Power BI dashboard was designed to provide a clear view of stock performance, sales activity, and inventory health. It includes interactive visuals and DAX-based KPIs to help users explore the data efficiently.

The dashboard was built across **two pages**, with **KPIs on each page**.

Page 1: Stock Overview

This page focuses on inventory monitoring and stock control.

KPIs included:

- Total Inventory Value
- Total Units in Stock
- Low Stock Items
- Out-of-Stock Products
- Stock Turnover Rate

Page 2: Sales and Supplier Performance

This page focuses on sales analysis and replenishment performance.

KPIs included:

- Total Sales Value
- Gross Profit
- Gross Margin %
- Reorder Compliance %
- Shrinkage Rate

Key Business KPIs

The project was designed to support the following business KPIs:

1. **Total Inventory Value** – Measures the total worth of stock on hand.
2. **Total Units in Stock** – Shows the overall stock quantity available.
3. **Low Stock Items** – Identifies products that need replenishment.
4. **Out-of-Stock Products** – Highlights products that are unavailable.
5. **Stock Turnover Rate** – Measures how quickly inventory is sold and replaced.
6. **Total Sales Value** – Tracks revenue generated from product sales.
7. **Gross Profit** – Measures profit after subtracting cost.
8. **Gross Margin %** – Shows profit efficiency as a percentage.
9. **Reorder Compliance %** – Evaluates how well stock replenishment is managed.
10. **Shrinkage Rate** – Measures loss due to discrepancies between expected and actual stock.

DAX Measures Used

The dashboard uses custom DAX measures to calculate business metrics. Examples include:

- Total Sales Value
- Total Cost
- Gross Profit
- Gross Margin %
- Total Units Sold
- Total Units in Stock
- Inventory Value
- Low Stock Items
- Out-of-Stock Items
- Shrinkage Rate

These measures were used to power cards, charts, and other visuals in the dashboard.

Project Features

- Realistic food retail stock dataset.
- Around 10,000 rows of data.
- Deliberate data quality issues for practice.
- SQL-based data cleaning and preparation.
- Power BI dashboard with interactive visuals.
- KPI-focused reporting for inventory and sales analysis.
- Suitable for portfolio, learning, and interview presentation.

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│
└── README.md
```

Learning Outcomes

This project demonstrates skills in:

- SQL data cleaning and transformation.
- Handling real-world data quality issues.
- Creating analytical datasets for BI reporting.
- Writing DAX measures in Power BI.
- Designing professional dashboards.
- Interpreting stock and sales KPIs for business decisions.

Future Improvements

Possible enhancements for this project include:

- Adding more store locations.
- Introducing seasonal sales patterns.
- Expanding supplier performance analysis.
- Adding forecast-based inventory alerts.
- Creating drill-through pages for product-level analysis.
- Implementing real-time data refresh logic.

Conclusion

This project shows how **SQL Server** and **Power BI** can be combined to build a practical stock management analytics solution for a food retail business. The dataset, cleaning process, and dashboard together provide a strong end-to-end example of data analysis, data preparation, and business intelligence reporting.